

Homework 1 - Web Science Intro

Due: Sunday, February 15, 2026 by 11:59pm

Read the entire assignment before starting.

Assignment

Write a report that contains the answers and *explains how you arrived at the answers* to the following questions. Before starting, review the [HW report guidelines](#). Name your report for this assignment `HW1-report` with the proper file extension.

Q1

Consider the "bow-tie" structure of the web in the Broder et al. paper ["Graph Structure in the Web"](#) that was described in Module 1.

Now consider the following links:

```
A --> B
B --> A
B --> C
C --> D
C --> G
D --> A
D --> H
E --> F
E --> O
F --> G
G --> C
H --> L
J --> N
K --> I
M --> A
N --> L
O --> J
```

Draw the resulting [directed graph](#) (either sketch on paper or use another tool) showing how the nodes are connected to each other and include an image in your report. This does not need to fit into the bow-tie type diagram, but should look more similar to the graph on slide 24 from [Module-01 Web-Science-Architecture](#).

For the graph, list the nodes (in alphabetical order) that are each of the following categories:

- SCC:

- IN:
- OUT:
- Tendrils:
 - indicate if the tendril is reachable from IN or can reach OUT
- Tubes:
 - explain how the nodes serve as tubes
- Disconnected:

Q2

Demonstrate that you know how to use `curl` and are familiar with the available options. Complete the following steps using https://www.cs.odu.edu/~mweigle/courses/cs532/ua_echo.php as the URI to request. Explain the results you get from each step.

a) First, load the webpage at the URI in your web browser. The result should show the "User-Agent" HTTP request header that your web browser sends to the web server. Take a screenshot to include in your report.

b) Use a single `curl` command with the appropriate options to do the following:

- request the URI
- show the HTTP response headers
- follow any redirects
- change the User-Agent HTTP request field to "CS432/532"

Take a screenshot of the `curl` command and options you used and the result of your execution to include in your report.

c) Use a single `curl` command with the appropriate options to do the following:

- request the URI
- follow any redirects
- change the User-Agent HTTP request field to "CS432/532"
- save the HTML output to a file

Take a screenshot of the `curl` command and options you used and the result of your execution to include in your report.

d) View the HTML output file that was produced by `curl` from part c in a web browser and take a screenshot to include in your report.

Q3

For some of our later assignments, you will be analyzing large numbers of webpages. The goal of this question is to write a Python program to gather URIs of webpages that have enough text so they will be useful later.

Your program must do the following:

- take the URI of a seed webpage as a command-line argument
- extract all the links from the page's HTML
- for each link, request the URI and use the `Content-Type` HTTP response header to determine if the link references an HTML file (`text/html`)
 - if it does, use the `Content-Length` HTTP response header to determine if it contains more than 1000 bytes
 - if it does, then print the final URI (after any redirects)

Use this program to collect at least 500 **unique** URIs of webpages that contain more than 1000 bytes. Save the list of URIs to a file to use in later assignments. The file must be uploaded to your GitHub repo.

Be aware of the need for a timeout. If you use the Python requests library, make sure to include the [timeout](#) parameter to your call to `get()`.

- Example: `requests.get(url, timeout=5) # 5 second timeout`

There are a couple ways you can use your program to gather the URIs:

- run the program multiple times with different seed webpages until you get to 500
- have your program randomly pick a URI that you've collected and use that as the new seed until you've collected 500 unique URIs

Answer-

Question one asks to graph the assignment listed for the functions of

A --> B
 B --> A
 B --> C
 C --> D
 C --> G
 D --> A
 D --> H
 E --> F
 E --> O
 F --> G
 G --> C
 H --> L
 J --> N
 K --> I
 M --> A
 N --> L
 O --> J

These values can be compared to the same scale found in the below graphing images by kumar et al.

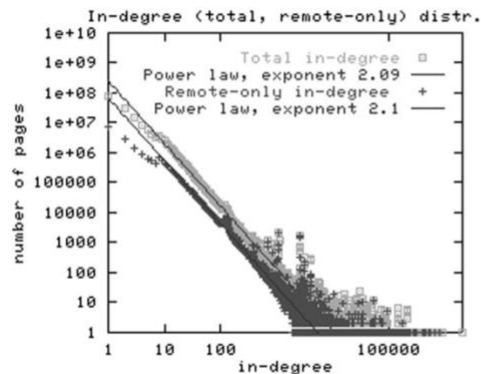


Fig. 1. In-degree distributions subscribe to the power law. The law also holds if only off-site (or 'remote-only') edges are considered.

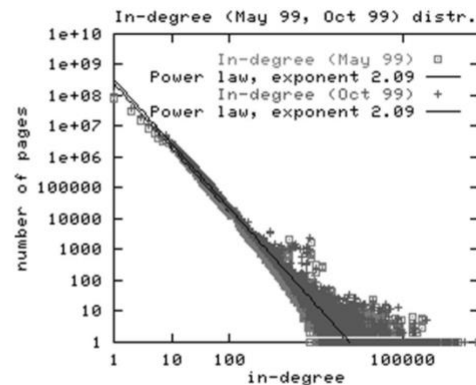


Fig. 3. In-degree distributions show a remarkable similarity over two crawls, run in May and October 1999. Each crawl counts well over 1 billion distinct edges of the Web graph.

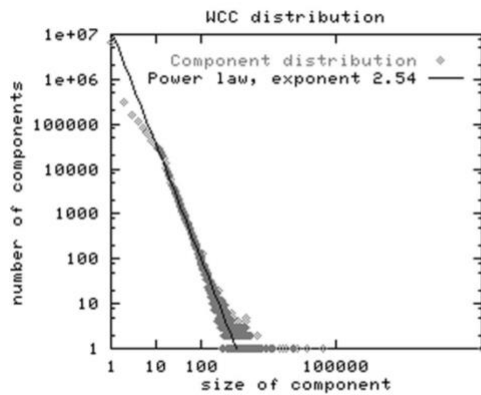


Fig. 5. Distribution of weakly connected components on the Web. The sizes of these components also follow a power law.

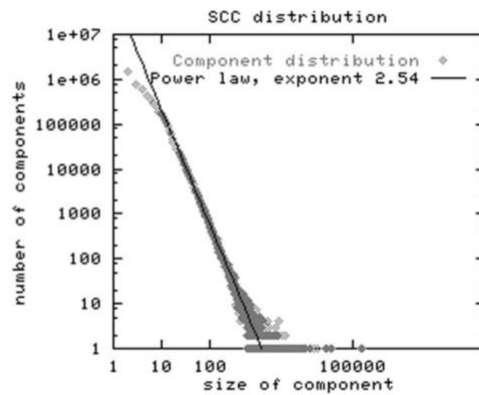


Fig. 6. Distribution of strongly connected components on the Web. The sizes of these components also follow a power law.

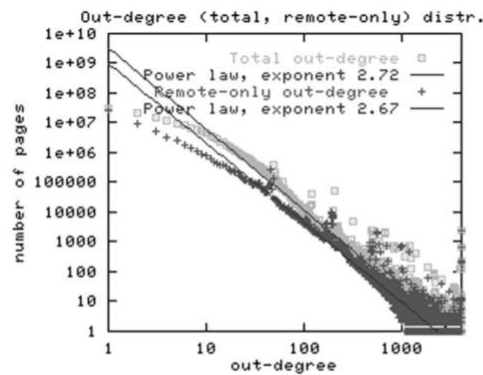


Fig. 2. Out-degree distributions subscribe to the power law. The law also holds if only off-site (or 'remote-only') edges are considered.

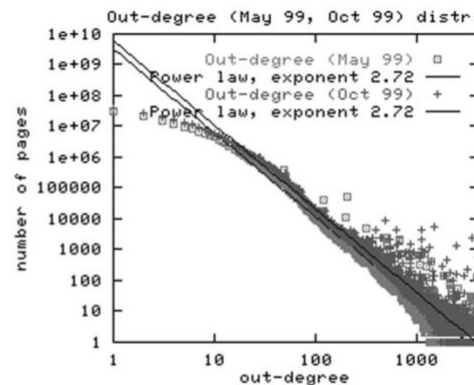
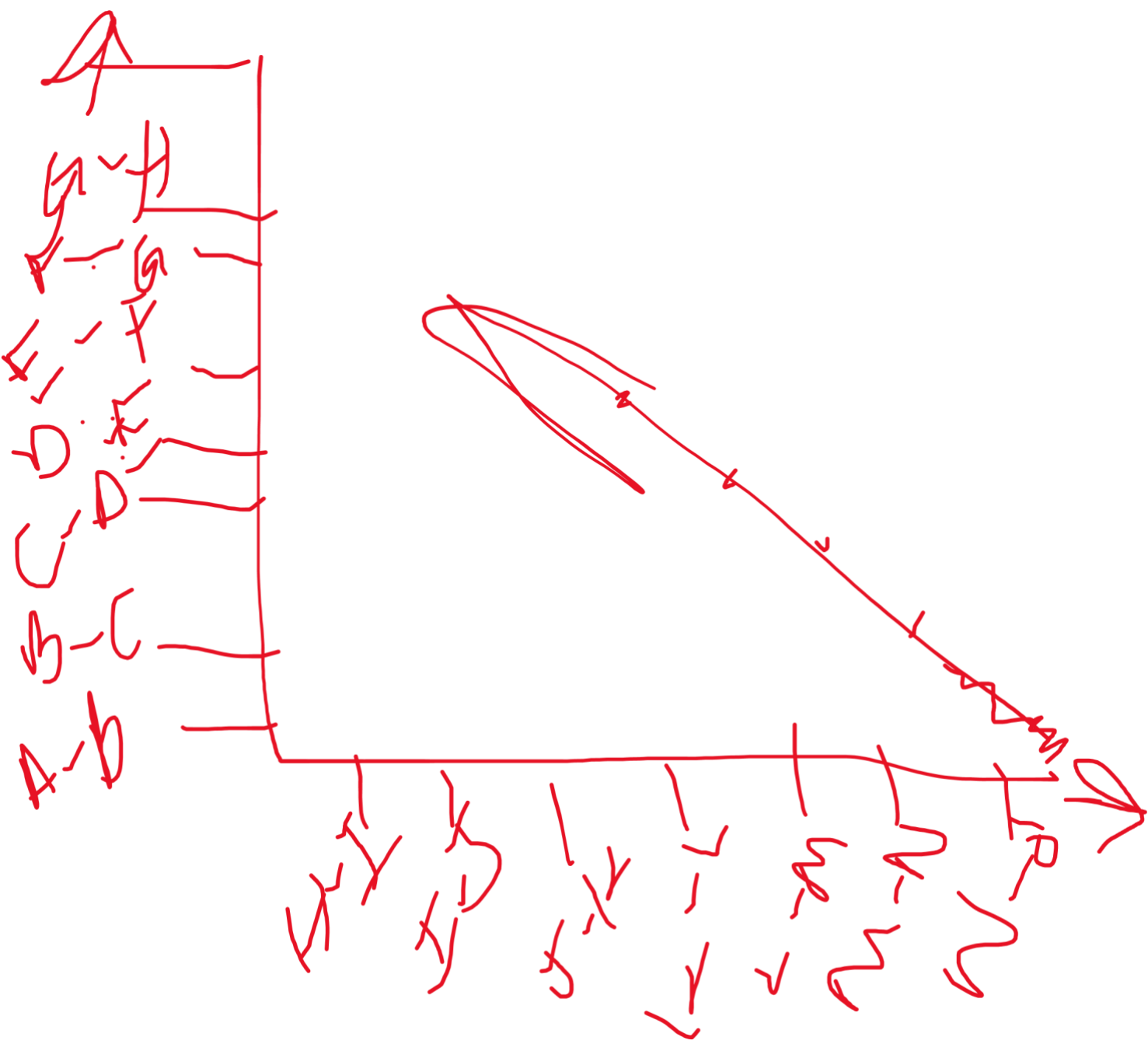


Fig. 4. Out-degree distributions show a remarkable similarity over two crawls, run in May and October 1999. Each crawl counts well over 1 billion distinct edges of the Web graph.

The graph created represents the in-degree, out-degree, and distributed functions to the solution.



In my graph above shows the upward progression of the values.

In the next question the assignment asks to code in python using https://www.cs.odu.edu/~mweigle/courses/cs532/ua_echo.php To retrieve curl and an input output for search, those images are located below in the image.

Branches

New branch

OverviewYoursActiveStaleAll

Q Search branches...

Default

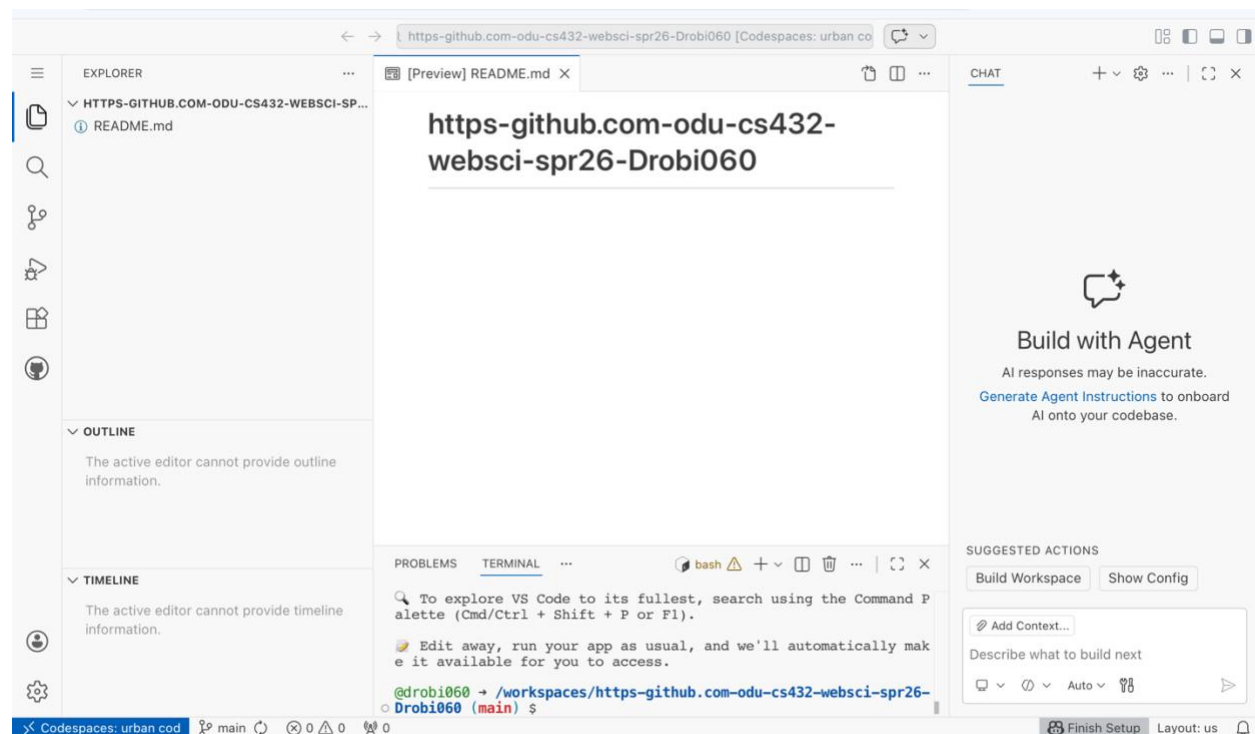
Branch	Updated	Check status	Behind	Ahead	Pull request
main	last week			Default	

Your branches

Branch	Updated	Check status	Behind	Ahead	Pull request
https://www.cs.odu.edu/mweigle/cou...	1 minute ago		0	0	

Active branches

Branch	Updated	Check status	Behind	Ahead	Pull request
https://www.cs.odu.edu/mweigle/cou...	1 minute ago		0	0	



The last question asks to present the python data of 500 url to represent a request and loaded webpage -without error. Shown below,

```
% python3 collect-webpages.py https://weiglemc.github.io/
```

```
https://weiglemc.github.io/
```

```
https://weiglemc.github.io/publications/
```

```
https://weiglemc.github.io/teaching/
```

```
https://weiglemc.github.io/students/
```

```
https://weiglemc.github.io/talks/
```

```
https://weiglemc.github.io/schedule/
```

```
https://weiglemc.github.io/cv/
```

```
https://weiglemc.github.io/contact/
```

```
https://www.odu.edu/
```

```
https://www.odu.edu/facultydevelopment/women-in-  
stem#tab9=3&done1612907281342
```

```
https://www.odu.edu/computer-science/academics/graduate/masters
```

```
https://www.odu.edu/computer-science/academics/graduate/phd
```

```
https://oduwsdl.github.io/
```

```
https://weiglemc.github.io/publications/recent
```

```
https://minerva.defense.gov/Research/Funded-Projects/Article/2957187/innovating-  
interdisciplinary-methods-for-hard-to-reach-environments/
```

```
https://oducsreu.github.io
```

```
https://www.odu.edu/computer-science
```

```
https://www.clemson.edu/cecas/departments/computing/
```


<https://cs.unc.edu/>

<https://www.unc.edu/>

<http://jekyllrb.com/>

<https://mademistakes.com/work/jekyll-themes/minimal-mistakes/>

% python3 collect-webpages.py <https://weiglemc.github.io/> 200

Need to collect 178 more URIs...

random seed: <https://cs.unc.edu/>

Need to collect 125 more URIs...

random seed: <https://cs.unc.edu/event/resume-workshop/>

Need to collect 121 more URIs...

random seed: <https://give.unc.edu/donate?p=COMP>

Need to collect 114 more URIs...

random seed: <https://cs.unc.edu/undergraduate/lecture-archive/>

Need to collect 110 more URIs...

random seed: <https://weiglemc.github.io/talks/>

Need to collect 89 more URIs...

random seed: <https://www.matchinggifts.com/unc/>

Need to collect 77 more URIs...

random seed: <https://cs.unc.edu/undergraduate/ug-advising/>

Need to collect 71 more URIs...

random seed: <https://cs.unc.edu/research/facilities/>

Need to collect 69 more URIs...

random seed: <https://cs.unc.edu/person/john-majikes/>

Need to collect 68 more URIs...

random seed: <https://weiglemc.github.io/students/>

Need to collect 54 more URIs...

random seed: <https://weiglemc.github.io/talks/2020-11-09-course>

Need to collect 54 more URIs...

random seed: <https://sigir.org/sigir2019/>

Need to collect 41 more URIs...

random seed: <https://cs.unc.edu/undergraduate/courses-for-non-majors/>

Need to collect 41 more URIs...

random seed: <https://www.loc.gov/programs/web-archiving/about-this-program/>

Need to collect 30 more URIs...

random seed: <https://sciences.sorbonne-universite.fr/>

Collected 200 unique URIs

<https://weiglemc.github.io/>

<https://weiglemc.github.io/publications/>

<https://weiglemc.github.io/teaching/>

<https://weiglemc.github.io/students/>

<https://weiglemc.github.io/talks/>

<https://weiglemc.github.io/schedule/>

<https://weiglemc.github.io/cv/>

<https://weiglemc.github.io/contact/>

...

R. Kumar, P. Raghavan, S. Rajagopalan and A. Tomkins, Extracting large scale knowledge bases from the web, in proc. VLDB, July 1999.